# Determine

Dumb glasses, a capable phone, and a calibrated warning about what is directly ahead of you.

**Architecture and prototype contract**

Status: proposed design. Nothing here is built, clinically validated, or cleared as a medical device. It does not replace a white cane, a guide dog or orientation-and-mobility training.

**WEARABLE**
Two dumb boards: an ESP32-CAM that streams JPEG, and a XIAO ESP32-C3 carrying the time-of-flight array, stereo bone-conduction cues and two haptic motors. No AI on the frame.

**PHONE**
Does all perception, decides which hazard to report, runs calibration and the user interface. Android first.

**USERS**
Blind and low-vision travellers, including people with unilateral or asymmetric hearing loss.

**TARGET**
HackTX 26, Oct 24–25 2026, UT Austin. Build on site within the event rules.

> **Pitch.** When you reverse a car, your view is limited, so the car gives you parking sensors. People who are actually blind or have low vision mostly do without a head-level equivalent. Determine is that sensor for a walker: it beeps faster as an obstacle gets closer, places the warning on the left, centre or right, and calibrates its loudness so both sides are heard equally, even with hearing loss in one ear.

## How to read this contract

This document fixes *interfaces and invariants*. It does not claim that the hardware works or that the design has been field-tested. **MUST**, **MUST NOT**, **SHOULD** and **MAY** state obligations on the proposed implementation only. Every claim carries a tag:

| Tag | Meaning |
|---|---|
| FACT | Read from a primary or vendor source, cited with [n]. Vendor statements describe the vendor's own product, not an independent check of Determine. |
| DESIGN | An engineering decision this document makes, with its rationale stated. |
| INFERENCE | Arithmetic or reasoning from facts, not measured. A gate must confirm it before anyone repeats it as a result. |
| PROPOSED | A starting threshold to tune in a pilot. It is not a standard, a regulation or a measurement. |
| GATE | Measured evidence required before a capability may be claimed or the next layer built. |

Vendor prices below are historical checks from 15–24 Oct 2026, excluding tax and shipping; the current bill of materials quotes no unverified total. The user reports the listed Houston Micro Center components in stock, but automated reads of Micro Center inventory returned HTTP 403 and no reservation or receipt was observed. The Mini ToF sensor and raw bone transducers require separate delivery unless an exact local source is secured. Layout follows the reference *MisaOS-Architecture.pdf*. Determine is an independent project.

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## Decisions at a glance

| Question | Decision | Why |
|---|---|---|
| Where does computing happen? | On the phone. Two dumb boards on the glasses capture, transmit and play back. DESIGN | Cheap, replaceable frame. The phone already has an NPU, a battery and accessibility settings. One place to update models. |
| Two cameras? | No, for v0. Use one RGB camera plus an 8×8 time-of-flight (ToF) array. DESIGN | Stereo depth needs synchronised capture and calibration. ToF measures distance whatever the obstacle class, which is what head-level hazards need. |
| Camera on the cue board? | No. A separate ESP32-CAM streams JPEG over the same hotspot. DESIGN | The XIAO ESP32-C3 has no camera, and the esp32-camera driver supports ESP32, S2 and S3 only [13][59]. A second Wi-Fi producer is the price of keeping the cue board simple. |
| Camera link | JPEG from the ESP32-CAM over the phone's Wi-Fi hotspot. USB UVC only after a per-phone check. DESIGN | BLE cannot carry useful video [15]. Android external-camera support depends on the phone maker [10]. |
| Urgent cue path | The phone sends a cue ID and the cue board plays a stored earcon and haptic pattern. No A2DP. DESIGN | A2DP typically adds 100–200 ms [16]. Reflex cues need to arrive in tens of milliseconds. |
| Conveying direction | Redundantly: calibrated audio side, haptic side, and an optional spoken word. Never stereo panning alone. DESIGN | Bone-conducted sound crosses the skull [26]. People with single-sided deafness cannot use level differences [28][29]. |
| Calibration | Per session, per side: detection threshold, then a loudness balance, with a hard ceiling. DESIGN | This is the user's core requirement. Coupling and hearing differ by person and by how the frame sits. |
| Cloud | Never in the hazard loop. Scene description is optional and labelled "may be wrong". DESIGN | Network loss must not silence warnings. A 2026 study found multimodal-LLM answers wrong 22.2% of the time [30]. |
| Meta and AWS | Meta DAT is an alternative frame source behind the same interface. AWS is user-requested description only: a Graviton2 Lambda API plus SmolVLM on Inferentia2 is the design target, untested (§7.3). DESIGN | Shows real integration without weakening the cost thesis or the offline invariant. |

CHAPTER 1

# Mission, guarantees and non-goals

## 1.1 The precise product

Determine is a **secondary electronic travel aid** (ETA). It adds a near-field warning layer, focused on chest- and head-level obstacles, on top of the traveller's primary aid. The 2026 *Scientific Reports* study defines primary aids as the white cane and guide dog, and ETAs as supplementary tools [2]. A long cane covers the ground and "does not protect users from obstacles at the upper body and head levels" [2]. In a survey of 307 blind travellers, 13% had a head-level accident at least monthly, 86% of head-level accidents happened outdoors (mostly tree branches), and 23% of head-level accidents had medical consequences [3]. FACT

The wearable is deliberately "dumb": it has sensors, radios and actuators, but it makes no decisions. The phone app is the processing unit. It decides what counts as a hazard, which single cue to play and how loud each side should be. DESIGN

## 1.2 Guarantees

1. **G1 Fail loud.** If the camera, ToF, link, phone app or wearable power fails, the system targets an audible or tactile "system unavailable" notification through a surviving output within ≤ 1 s PROPOSED. If both boards lose power, the phone uses its own speaker and vibration; the dead frame cannot play an earcon. Silence means "nothing to report *from a working system*", never "path clear".
2. **G2 Unknown is not clear.** An empty detection list never produces a reassurance cue. Determine has no "all clear" sound.
3. **G3 Offline hazard loop.** Capture → perception → decision → cue runs with no internet connection.
4. **G4 Calibrated, capped output.** Per-side gain comes from the wearer's own calibration and is clamped by a ceiling that no setting can exceed.
5. **G5 Redundant direction.** Every directional cue is carried by at least two channels the wearer has confirmed during calibration that they can perceive.
6. **G6 Bounded vocabulary.** Hazard cues come from a small, learnable set (Chapter 8). Free-form speech never pre-empts a hazard cue.
7. **G7 Privacy by default.** Frames stay in phone memory. Images are not retained or uploaded unless the user explicitly turns on a separate feature.

## 1.3 Explicit non-goals

| Not promised | Reason |
|---|---|
| Replacing the cane, guide dog or O&M training | Primary aids provide ground contact and legal recognition. ETAs supplement them [1][2]. |
| Drop-offs, curbs, stairs down, holes | Forward-facing head-level sensing does not see the ground ahead reliably. No claim without per-class test data. |
| Street crossing or traffic judgement | Safety-critical. Out of scope. |
| Glass doors, thin wires, wet or very dark surfaces | Known failure modes for camera and ToF sensing, listed as "cannot see" in the app. |
| Restoring directional hearing | Bone conduction and rerouting do not restore localisation in single-sided deafness [28]. |

| Not promised | Reason |
|---|---|
| A certified sound-exposure dose from bone conduction | No verified standard maps bone-conduction transducer output to air-conduction dose limits. The prototype caps output but cannot certify it. |
| Unsupervised public trials during HackTX | Demos run on a staged indoor course with a sighted spotter. |

## 1.4 Claim classes

| Class | Meaning | Example |
|---|---|---|
| **C0** | Measured on the prototype, with the log kept | "p95 capture-to-cue latency was 212 ms over 120 trials." Only after the test exists. |
| **C1** | Vendor or primary-source fact about a part | "The VL53L5CX module has 8×8 zones and a 4 m range" [20]. |
| **C2** | Design target | "Unavailable cue within 1 s." |
| **C3** | Research direction | "Deliver cues to LE Audio hearing aids." |

The pitch, slides and Devpost entry **MUST** label class C2 and C3 statements as such. Presenting a C2 target as a C0 result is a release blocker.

CHAPTER 2

# Evidence base: travellers, hearing and cues

## 2.1 What a real ETA trial shows

Pittet et al. (2026) tested 13 blind and low-vision participants on an indoor corridor with floor, waist and head-level obstacles. They compared cane alone with cane plus NOA (a camera-based harness) and cane plus BuzzClip (ultrasonic) [2]. FACT

| Measure (mean) | Cane only | Cane + NOA | Cane + BuzzClip |
|---|---|---|---|
| Body contacts with obstacles | 2.92 | 1.62 | 3.54 |
| Cane contacts | 4.69 | 2.62 | 3.15 |

What this means for Determine:

- **Precision beats volume.** Participants rated NOA higher on precision and on upper-body detection. The BuzzClip's vibrations were criticised as too weak and too frequent, and gave no improvement in incidents [2].
- **Learnability matters.** 4 of 13 participants had trouble understanding NOA's spatialised warning sounds [2]. Determine therefore uses a small set of cues, trained during onboarding, with direction carried redundantly.
- **Short, precise beats generative.** For finding objects, every participant preferred concise output such as "Person to your left at 11 o'clock, 3 metres" from a local deep-learning model over generative-AI descriptions [2].
- **Test with the real population.** Of 140 ETA studies reviewed, only 29% included blind or low-vision participants [2]. A blindfolded sighted tester does not validate Determine.

## 2.2 The glasses premise, without stereotypes

Some blind people wear dark glasses, but many do not. Blindness is a spectrum, and many legally blind people use their remaining vision. Determine therefore offers clear, tinted and clip-on configurations, and its sensors mount on the frame front so that lens tint does not affect sensing. DESIGN A pair of glasses is a natural place for the hardware: it sits at eye level, it points where the head points, and it already rests on the temples, where bone conduction and haptics make contact.

## 2.3 Hearing: two different problems

### Asymmetric hearing loss

Both ears hear, but one hears less. Per-side gain can equalise *perceived* loudness, which is exactly what the user asked for. Calibration is the right tool here. INFERENCE

### Single-sided deafness (SSD)

One cochlea does no useful work. Every left or right channel ends up at the same working cochlea. In 28 unilaterally deaf subjects, sound crossing the skull lost only about 0–10 dB, depending on frequency, with roughly 40 dB variation between people [26]. Left/right audio therefore cannot carry direction, so haptics and spoken words must. FACT → DESIGN

- Bone-conduction headsets lateralise sound much like headphones in a small study. Even a 20 dB level difference gave only about 75% correct lateralisation [27]. Stereo bone conduction is useful, but it is not reliable enough to be the only carrier of direction. FACT

- In one study, a contralateral-routing bone-conduction device did not improve localisation for people with single-sided deafness [28]. Monaural listeners rely heavily on head-shadow level cues [29]. FACT
- Audio played through bone-conduction headphones increased localisation errors for real environmental sounds [52]. Determine keeps cues short and sparse so it does not mask traffic and voices. FACT → DESIGN

### 2.4 Auditory-display findings used in Chapter 8

- **Earcon guidelines** (Brewster et al.): vary timbre, register and rhythm, and use musical timbres rather than sine waves. Make rhythms differ substantially, including in note count. Keep pitch roughly within 125–150 Hz to 5 kHz. Notes should last at least 0.0825 s, at most about 6 notes per second, with about 0.1 s between serial earcons. Keep intensity 10–20 dB above threshold. Do not rely on register or intensity alone for absolute identification [24]. FACT
- **Spearcons**, which are sped-up spoken words, beat earcons and auditory icons on menu-navigation efficiency and accuracy, and are learned about as fast as speech [25]. FACT
- **Safe listening:** WHO lists 80 dB as safe for up to 40 hours a week and 90 dB for only 4 hours. It recommends keeping devices at or below 60% of maximum volume [4]. The WHO-ITU standard defines exposure modes for adults and sensitive users [5]. FACT

CHAPTER 3

# Landscape and positioning

| Product | Form | Relevant capability | Price (source, 24 Oct 2026) | Limitation relative to Determine's goal |
|---|---|---|---|---|
| Ray-Ban Meta (Gen 3) | Camera and open-ear speaker glasses | Meta AI scene description, Be My Eyes volunteer calls. Developer access via Wearables Device Access Toolkit (DAT) 1.0 [6] | From $449 [42] | AFB calls it an accessibility tool "only by coincidence" and reports hallucinated object positions [9]. Air-conduction speakers, not bone conduction. No haptics. Gen 3 is not named in the DAT supported-device list [6] |
| Envision Ally Solos | Camera glasses, phone-tethered | Reading, scene description, faces | $699 shop listing [43] | Description-focused, not near-field hazard cues |
| OrCam MyEye 3 Pro | 22.5 g clip-on camera | Reading, faces, products | $4,250 reseller [44] | Reading-centric. Not a navigation device |
| WeWALK Smart Cane 2 | Smart cane handle | Chest and head-level obstacle detection with vibration, navigation app | Not on the static official page [45] | Occupies the cane hand. Closest neighbour in function |
| biped NOA | Shoulder harness, three depth cameras | Spatialised obstacle audio sent to bone-conduction headphones [2] | $4,990 reseller [46] | 8 of 13 trial users found it heavy or bulky [2]. Expensive |
| .lumen glasses | Head-worn, forehead haptics | Haptic path guidance | €9,999 [47] | Ships only in Romania per its order page. Claims such as "better than the guide dog" are unvalidated marketing |
| NaviLens | Markers plus a free app | Long-range marker reading in transit systems | Free [48] | Needs installed infrastructure. Wayfinding, not obstacle avoidance |
| Seeing AI / Lookout | Phone apps | Text, scene and object description | Free [49][50] | Handheld. Lookout's Explore mode is beta and "less accurate" [50] |
| Microsoft Soundscape | Retired spatial-audio app | 3D audio beacons | Open-sourced [51] | Microsoft says it is "not … for use as a mobility aid" [51] |

## 3.1 Where Determine sits

None of the products found combines all four of the following: a low-cost glasses form, phone-side perception, bone-conduction *and* haptic output, and explicit calibration for asymmetric hearing. No ETA product or paper found addresses unilateral hearing loss in its cue design. That is an inference from absence of evidence, not a verified gap. INFERENCE

**Not "AI glasses for the blind"**

That framing is crowded and invites overclaiming. Determine is a *parking sensor for walking*, whose output follows a defined set of rules.

**Commodity hardware, unpriced build**

The phone does perception; the frame uses two commodity boards. Only $88.35 of line items have quoted prices, including out-of-stock transducers. Board packs, drive extras and power remain unpriced (Chapter 12). Do not quote a frame total yet.

**Hearing-aware by construction**

Calibration, a mono-safe mode and haptic direction are core features, not settings bolted on later.

**Claims to avoid in the pitch:** "replaces the cane", "detects stairs, curbs or traffic", "makes navigation safe", "restores hearing", "first obstacle sensor for the blind" (WeWALK and NOA already exist), "uses Auracast" (unless it was actually built and demonstrated).

CHAPTER 4

# System architecture and seams

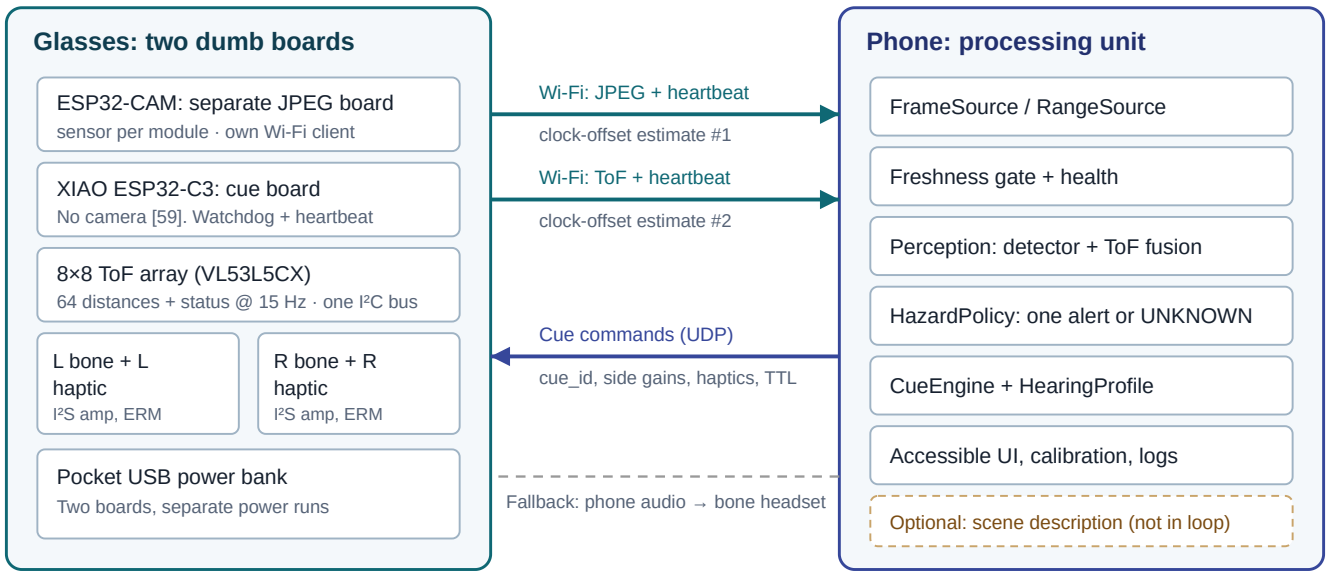



Figure 4.1. Topology: two independent Wi-Fi producers and no board-to-board bus. The phone is the only integrator — it aligns each stream by that board's estimated clock offset and receive age, and discards stale input. The wearable never decides what is hazardous. Cue commands carry a time-to-live (TTL), so a stale command is dropped rather than played late.

## 4.1 Module seams

Each seam exists only where at least two implementations are real or planned, following the "two adapters means a real seam" rule. DESIGN

| Module | Interface | Adapters | Invariants and error modes |
|---|---|---|---|
| **FrameSource** | `next(): Frame{seq, tCapture, w, h, jpeg}` or `Unavailable(reason)` | ESP32-CAM Wi-Fi · USB UVC · Meta DAT · phone camera (Tier 0) | Sequence numbers increase monotonically. Clock offset is estimated per board by PING/PONG; receive age is tracked. Stale input is discarded. Disconnection is explicit, never an empty frame. |
| **RangeSource** | `next(): RangeGrid{seq, t, mm[64], status[64]}` | VL53L5CX on the XIAO ESP32-C3 · none (camera only) | An invalid zone means UNKNOWN, never "far". Same per-board offset and staleness rules as frames. |
| **Perception** | `evaluate(frame, range?): Observations` | MediaPipe detector · plus optional depth model | Every observation carries its source age and confidence. No metric distance from bounding-box size alone. |
| **HazardPolicy** | `select(obs, motion): Alert \| Unknown \| Quiet` | Single rule-based implementation (internal seam for tests) | At most one alert at a time. Entry and exit use hysteresis. Never emits "clear". |

| Module | Interface | Adapters | Invariants and error modes |
|---|---|---|---|
| **CueSink** | `present(alert, profile): Ack \| Fault` | Glasses playback over UDP · phone audio to headset · phone vibration (non-directional) | Gains are clamped by the profile ceiling. A missing acknowledgement escalates to UNAVAILABLE. |
| **HearingProfile** | `calibrate(sink): Profile` · `gains(cue): (gL, gR)` | One per output device | Calibration is timestamped and re-run each session. A side outside the ceiling is marked unusable. |

## 4.2 Runtime state machine

```
STARTING ──(frames fresh ∧ ToF ok ∧ cue ack)──▶ READY
READY ──(ToF invalid ∨ low light ∨ confidence low)──▶ DEGRADED   # announce once; only supported claims
READY/DEGRADED ──(frame age > 500 ms ∨ no heartbeat > 500 ms from either board ∨ cue NACK ∨ phone
battery critical)──▶ UNAVAILABLE
UNAVAILABLE: play "system off" on any surviving cue output + phone speaker/vibration; suppress all
obstacle cues; retry link
any ──(user pause)──▶ PAUSED   # explicit, spoken confirmation; never automatic
```

The 500 ms frame-age and per-board heartbeat timeouts are **PROPOSED**; heartbeat interval targets 100 ms. They leave time inside the untested ≤ 1 s fail-loud goal and must be tuned at Gate G7. A bank without telemetry cannot report low charge before its outputs die.

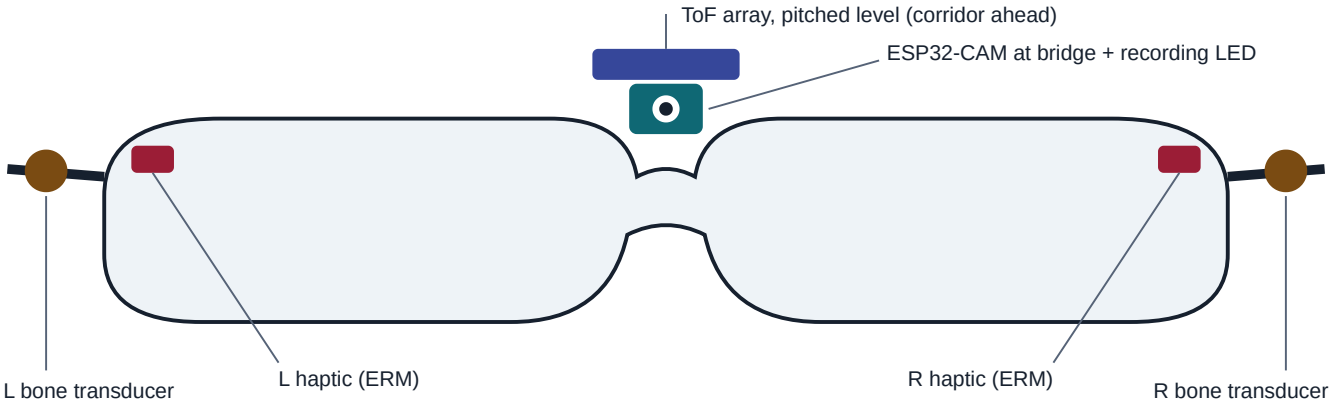
## 4.3 Wire contract (v0)

```
ESP32-CAM → phone (UDP/TCP over hotspot; little-endian; versioned)
  FRAME   {ver:u8, seq:u32, t_cap_us:u64, w:u16, h:u16, len:u32, jpeg[len]}
  BEAT    {ver, seq, t_us, faults:u16}
XIAO ESP32-C3 → phone
  RANGE   {ver, seq, t_us, mm:u16[64], status:u8[64]}
  BEAT    {ver, seq, t_us, faults:u16}
Phone ↔ each board (clock estimation; the camera board accepts no cue commands)
  PING    {t_phone_us}  →  PONG {t_phone_us, t_board_us}
Phone → XIAO ESP32-C3
  CUE     {ver, seq, t_issued_phone_us:u64, cue_id:u8, side:{L,R,BOTH},
           gain_l_db:i8, gain_r_db:i8, haptic:{pattern:u8, l:u8, r:u8}, ttl_ms:u16}
  PROFILE {ver, ceiling_db:i8, offsets_db[bands][2]}
```

There is no board-to-board bus: the phone is the only integrator. The phone estimates a separate clock offset for each board, timestamps receipt, and discards `FRAME` or `RANGE` whose offset-corrected capture time or receive age exceeds the freshness bound. The cue board enforces the gain ceiling independently. It compares `t_issued_phone_us + ttl_ms` against its offset-corrected current time, dropping a delayed `CUE` rather than resetting its lifetime on arrival. The C3's watchdog plays its system-off cue when phone PINGs stop while the board is powered; the phone uses its own speaker/vibration if a board disappears. Clock uncertainty, command expiry and both failure directions must be tested in Gates G6–G7. Power-bank battery percentage is not reported by either board unless external sensing is added; loss of its power is detected by missing heartbeats on the phone. **DESIGN**

CHAPTER 5

# Wearable hardware




Only the camera and Mini ToF face forward; cue board, amps and drivers sit on a rear strap. Power bank stays in a pocket.

Figure 5.1. Proposed compact headband layout, not finished eyeglasses. The single Mini ToF board is 12.7 × 25.4 mm [20]; no three-board ToF cluster. Weight is kept off the nose. Fit, cable routing and comfort remain untested.

## 5.1 Sensing options

| Configuration | What it gives | Cost and risk | v0? |
|---|---|---|---|
| Mono RGB only | Object classes and bearing (left/centre/right) from image position | No distance for arbitrary obstacles. Common detectors cover objects such as people, chairs and cars, but not branches or poles INFERENCE | Fallback |
| **Mono RGB + 8×8 ToF** | Class-agnostic distance in a forward corridor, plus semantics from the camera. VL53L5CX module: 4 m, 63° diagonal FOV, 15 Hz at 8×8, Class 1 940 nm laser [20] | $25.95 [20]. Loads ~90 KB of firmware over I²C at every boot [20]. Behaviour in sunlight and against glass is unmeasured | **Yes** |
| Dual RGB stereo | Dense depth after calibration | Needs synchronised exposure and stereo rectification. The ESP32-CAM carries one sensor and the C3 has no camera interface at all [13][59], so this needs two camera MCUs or a stereo module | No |
| Mono + learned depth | Relative depth. Depth Anything V2-Small is Apache-2.0, while its larger variants are non-commercial [23] | No metric scale unless anchored to the ToF. Phone compute load | Stretch |
| Stereo depth module (e.g. OAK-D Lite) | On-device depth | Expensive, multi-watt, harder on Android (secondary source only) | No |
| Meta Ray-Ban via DAT | Glasses camera stream up to 720×1280 at 2–30 fps over Bluetooth [6][58] | No ToF or haptics. Latency undocumented. Requires owning the glasses. Gen 3 not listed [6] | Optional adapter |

**Why ToF instead of a second camera?** Head-level hazards such as branches, sign edges and open cabinet doors are rarely detector classes. The hazard decision must not depend on recognising the object, and a ToF array measures distance regardless of what the object is. The camera adds meaning ("person", "door") and sharper bearing. DESIGN

## 5.2 Wearable board choice (two-board revision, 24 Oct 2026)

The XIAO ESP32-S3 Sense is not obtainable for this build, so the wearable splits into two boards that share no bus: a **cue board** (ToF over I²C, I²S stereo playback, two haptic outputs) and a separate **camera board** that only streams JPEG. Both join the phone's hotspot as independent clients (Chapter 4). DESIGN

| Board | Relevant facts | Verdict |
|---|---|---|
| **Seeed XIAO ESP32-C3 (3-pack)** | RISC-V single core, 400 KB SRAM and 4 MB flash, Wi-Fi b/g/n + BLE 5, exactly one I²C and one SPI bus, 11 GPIO, 21×17.8 mm, no integrated camera [59]. Vendor-typical 75 mA with Wi-Fi active [59] | **Selected: cue board.** Micro Center Houston SKU 069005; user reports the listing in stock, but no independently verified shelf count or price. One board used, two spares |
| **ESP32-CAM (classic ESP32)** | The esp32-camera driver lists ESP32, S2 and S3 as supported SoCs [13]. Practical streaming needs PSRAM: the driver requires it for anything above CIF-or-lower JPEG [13] | **Selected: camera board.** Inland pair, Micro Center Houston SKU 221739; user reports stock. The actual module's PSRAM and sensor must pass Gate G0 before use |
| Seeed XIAO ESP32-S3 Sense | 8 MB PSRAM, camera board (OV3660 or OV2640 since 2025-06-30), microphone, one DVP camera interface [12] [14] | **Not selected:** unavailable to this build. Documented single-board alternative for a later revision ($13.99 online when checked 24 Oct [14]) |
| Raspberry Pi Zero 2 W | Quad A53, CSI-2, H.264 1080p30 encode, Wi-Fi and BT 4.2, 5 V/2.5 A supply spec [53] | Overkill for a "dumb" frame. Power and boot time. Out of stock at Adafruit |
| Raspberry Pi Pico 2 W | RP2350, 520 KB SRAM, no camera port [54] | Cannot carry the camera role |
| ESP32-P4 | USB 2.0 high-speed, MIPI-CSI with ISP, H.264 up to 1080p30, but no radio [55] | Roadmap: wired high-speed or Wi-Fi via a companion chip |

## 5.3 Output hardware

- **Stereo bone conduction, played on the glasses.** Two MAX98357A I²S amplifiers share one I²S bus. Select left on one amp and right on the other: the default mixes both channels [36]. Each drives one 8 Ω / 1 W Adafruit 1674 transducer. Adafruit's own store was out of stock on 18 Oct, but DigiKey lists the same item (1528-1948-ND); its displayed inventory is not a checkout-confirmed delivery date. Limit output below the transducer's 1 W continuous rating, and cover each skin-facing transducer and wire joint against sweat [36]. FACT → DESIGN
- **Haptics.** Two ERM disc motors, one per side, each driven by a DRV2605L. Both drivers have the same I²C address, so the selected TCA9548A mux isolates them on separate channels; the Mini ToF may use the upstream bus [59]. The Micro Center driver listing may differ in connector style from Adafruit's current STEMMA QT revision: inspect it and wire its header as supplied. A MOSFET-PWM revision would require a different BOM and is not this build. DESIGN
- **Fallback audio.** A commercial bone-conduction headset paired to the phone, such as Shokz OpenMove at a $79.95 list price (variants showed sold out) or OpenRun at $129.95 [37]. Classic Bluetooth adds latency (§6.2), so in this mode haptics carry the urgent cue.

## 5.4 Power

The revision uses two Wi-Fi boards and a single compact front-facing Mini ToF board, not three 41 × 25 mm Modulino Distance boards. The Mini measures 12.7 × 25.4 mm, retains the 8×8 grid and 4 m vendor range [20], and is the justified shipped exception to Houston-first sourcing. The ESP32-CAM and ToF face forward; the C3, amplifiers, haptic drivers and power distribution sit toward the back of a headband. The power bank stays in a pocket, feeding the C3 and a separate 5 V distribution branch for the camera, amps and a 3.3 V LM3671 peripheral

regulator. All grounds needed for shared signals are common. The board-fit, thermal load and simultaneous output current are unmeasured; Gate G0 checks power and mounting before any wearable claim. DESIGN

GPIO budget: the C3 has one hardware I²C bus [59] for the Mini ToF and TCA9548A; the two DRV2605Ls sit on separate mux channels. The two amplifiers share three I²S signals. Fix the complete pin map, status-LED GPIO and channel-select resistor values before soldering; do not power motors or amps from an unmeasured C3 3V3 output. GATE G0

CHAPTER 6

# Links: Wi-Fi, USB, BLE, LE Audio, Auracast

## 6.1 Camera and sensor transport

| Link | Facts | Use in Determine |
|---|---|---|
| **Wi-Fi to the phone hotspot** | Both wearable boards join the hotspot as stations: the ESP32-CAM streams JPEG and the C3 sends ToF grids and acknowledgements. esp32-camera recommends JPEG, because YUV and RGB strain PSRAM bandwidth when Wi-Fi is on [13]. Both boards and the phone share one hotspot channel's airtime | **Primary.** QVGA–VGA JPEG at 10–15 fps target. Joining the phone's hotspot keeps the phone's cellular internet, which joining a glasses-hosted access point would not. INFERENCE Effective throughput with two stations is unmeasured; Gate G1 measures it |
| USB-C UVC | Android supports UVC webcams through Camera2, but only if the device maker enables the external camera provider. AOSP says it "isn't designed" for high-speed streaming or AR, and its sample configuration bounds 640×480 at 30 fps [10] | Secondary, only after a per-phone check with a UVC camera. A user-space UVC library is a further fallback |
| BLE GATT | ESP32-to-ESP32 throughput is about 0.73 Mbit/s on the 1M PHY and about 1.4 Mbit/s on the 2M PHY. Phone throughput is lower and undocumented [15] | Provisioning, health, and possibly cues. Not video |
| Meta DAT | The camera stream degrades resolution first, then frame rate, and does not drop below 15 fps. A2DP and HFP audio are mutually exclusive [6][7] | Alternative FrameSource for teams that own supported glasses |

## 6.2 Audio paths and latency

| Path | Latency facts | Role |
|---|---|---|
| Phone → Classic BT headset (A2DP) | Typically 100–200 ms [16] | Fallback and spoken detail only |
| Phone → LE Audio unicast (LC3) | LC3 uses 10 ms frames. Best case is "just over 20 ms" end to end. Every receiver must support a 40 ms presentation delay, and hearing-aid live use stays at ≤ 20 ms [16]. Android has had LE Audio since Android 13 [19] | Roadmap: deliver cues to LE Audio hearing aids and earbuds |
| Auracast broadcast | One-to-many, connectionless. The transmitter never knows who is listening [17]. Android support is limited to some devices (Pixel 8 and later, Galaxy S23 and later, some Xiaomi) [18] | Demo extra: judges listen on their own Auracast earbuds through an encrypted stream. Never the personal safety path |
| **Phone → cue board cue ID (UDP) → stored earcon** | LAN hop plus I²S playback. Unmeasured | **Primary urgent path.** True stereo, per-side gain applied on the cue board |

> **Why the glasses play the earcons.** The "dumb glasses" principle is about where decisions are made, not about refusing to store a 2-second sound bank. The phone decides *which* cue to play and *how loud* each side is. The glasses play it, which removes 100–200 ms of A2DP latency and the cost of a headset. DESIGN

### 6.3 End-to-end latency budget

| Stage | Budget | Basis |
|---|---|---|
| Exposure + on-chip JPEG encode | ≤ 60 ms | PROPOSED |
| Wi-Fi transfer (~30 KB frame) | ≤ 40 ms | ~1 MB/s effective throughput assumed for one station; the C3 shares the hotspot's airtime, so this is unmeasured for two producers. INFERENCE |
| Decode + detector + fusion on phone | ≤ 80 ms | Benchmark the actual model on the actual phone |
| Policy + cue dispatch + playback start | ≤ 40 ms | UDP on LAN. INFERENCE |
| **Capture-to-cue p95 target** | **≤ 250 ms** | **Target kept from the single-board design; unmeasured with two Wi-Fi producers. The ToF path skips the camera stages and should be faster** |

At a walking speed of 1.4 m/s, every 100 ms of latency is about 14 cm of travel. INFERENCE So 250 ms is about 35 cm before human reaction time is even counted. Warning distances in the hazard policy have to include both.

CHAPTER 7

# Perception and hazard policy

## 7.1 Pipeline

1. **Ingest and freshness gate.** Two independent producers: the ESP32-CAM's frames and the cue board's ToF grids each carry their board's clock, a per-board offset estimated by PING/PONG, and the phone's receive timestamp. Drop any input whose offset-corrected capture time or receive age exceeds the freshness bound. Every stage records a timestamp, for Gate G6.
2. **Range corridor (class-agnostic).** Split the 8×8 ToF grid into a *corridor* (central columns, head-to-chest rows) and left and right flanks. Use the minimum valid distance per region. Invalid zones count as UNKNOWN.
3. **Semantics.** MediaPipe Object Detector in `LIVE_STREAM` mode with a category allowlist and a score threshold [21], running on the LiteRT runtime [22]. Classes are used for *labels* ("person", "door", "chair") and for bearing, never as the only trigger for a near-field hazard.
4. **Fusion.** Match detections to ToF regions by bearing. If a ToF return lies in the corridor and no detection matches, report it as a generic "obstacle".
5. **Proximity band.** FAR / MID / NEAR come from ToF distance when it is valid. Without ToF, use only *relative* cues such as bounding-box growth over time, and never speak metres.
6. **Policy.** Pick at most one alert, with persistence and hysteresis (below).

## 7.2 Hazard policy (v0 rules)

```
inputs: corridor_min_mm, left_min_mm, right_min_mm (valid flags), detections, head_yaw_rate
band(d): NEAR if d < 1000; MID if d < 2000; FAR if d < 3000; else NONE      # PROPOSED mm
enter alert:  same region ∧ band ≥ FAR for ≥ 3 consecutive ToF samples (~200 ms @15 Hz)
exit alert:   band NONE for ≥ 5 samples, or region changes (re-enter with new cue)
priority:     NEAR corridor > NEAR flank > MID corridor > MID flank > FAR corridor
rate limit:   ≤ 1 alert onset / 700 ms; repeat cadence set by band (Chapter 8)
suppress:     head_yaw_rate high (scanning) → hold FAR cues; NEAR always passes
label:        matched detection class if confident, else "obstacle"
never:        output CLEAR; speak distance without valid ToF; let description pre-empt NEAR
```

The bands are PROPOSED. With 250 ms of system latency and about 1 s of human reaction at 1.4 m/s, a hazard reported at 1.0 m leaves very little margin. That is why the FAR and MID bands exist, and why the NEAR band is a last-moment warning, not the main mechanism. INFERENCE White-cane mode MAY suppress low-zone returns that the cane already covers. The Ara wearable uses a similar mode [1].

## 7.3 Optional scene description (Describe loop)

The user asks with a long press or voice: "What's ahead?" — never automatically. If the user has opted into cloud description and the service is available, the phone redacts faces and sensitive detail before upload; imperfect redaction is a privacy risk, so upload happens only on the user's explicit choice. A small arm64 Lambda API on Graviton2 [65] holds the prompt — two sentences, surroundings first, no guesses about people — and a small vision-language model describes the image: SmolVLM via Hugging Face Optimum Neuron on one Inf2 instance [62] [63]. The model family is documented for Inf2, but this deployment is **untested by us**. Trainium2 is the documented alternative [60][61][64], and Bedrock Nova Lite is a fallback that does not guarantee the silicon [66]; new AWS accounts start with zero Inf/Trn quota, so accelerator access must be requested early [67]. If no description service is ready, the phone says description is unavailable; it does not improvise an untested local model. A returned answer is prefixed "may be wrong". MID or NEAR hazards interrupt it, and it is inactive while the system is UNAVAILABLE. Response time is unmeasured; the design plans for seconds, not milliseconds. DESIGN

### 7.4 "Cannot see" list (spoken during onboarding)

Glass doors and windows · thin wires and ropes · drop-offs, curbs and stairs going down · objects outside the forward field of view · fast-moving vehicles · glare, darkness, rain and fog · anything beyond about 3 m.

CHAPTER 8

# The cue language

The requirement is a sound that is granular, easy to identify and level-aware. Determine treats the cue as a small grammar with four independent dimensions, so that no single dimension, and especially not loudness alone, has to carry everything. Brewster's guidelines warn against identification by register or intensity alone [24]. FACT → DESIGN

| Dimension | Encoded by | Values | Rationale |
|---|---|---|---|
| **What** (family) | Timbre + rhythm motif | Obstacle: short wood-block double. Person: two-tone marimba. System: descending three-note. Ready: single soft tick, once | Brewster: timbre and rhythm are the strongest discriminators [24] |
| **Where** (side) | Calibrated side + matching haptic + optional spearcon | Left · Centre (both sides together, both motors) · Right | Redundant, so it survives single-sided deafness and the skull-crossing of bone-conducted sound [26][28] |
| **How close** | Repetition rate (parking-sensor metaphor) | FAR 1 per s · MID 2.5 per s · NEAR 5 per s plus urgent motif | Matches the car-sensor mental model. Stays within about 6 notes per second [24] |
| **How urgent** | Limited level step + haptic intensity | Base at calibrated sensation level. NEAR at +6 dB *within the ceiling* | Level supports urgency but is never the only signal. Stays 10–20 dB above threshold [24] |

## 8.1 Timing grammar

```
note ≥ 83 ms (motif notes)   gap between serial cues ≥ 100 ms   max 6 notes/s      [24]
pitch range 300 Hz – 4 kHz  (inside Brewster 125 Hz–5 kHz and inside transducer response)
FAR:  ■■ ............ ■■ ............  (1/s)
MID:  ■■ ... ■■ ... ■■ ...            (2.5/s)
NEAR: ■■■ ■■■ ■■■ + haptic buzz       (5/s, urgent motif, +6 dB within ceiling)
SYSTEM OFF: ■ □ □ descending, repeated every 5 s until resolved, plus phone vibration
```

The lower edge of the pitch range assumes the bone transducer's 300 Hz–19 kHz response [36]. Recheck it against whatever transducer is actually fitted.

## 8.2 Spoken layer

Spearcons are sped-up words such as "left", "door" or "person". They are optional and follow the earcon, never replace it. They are on by default during training and become opt-in once the wearer is fluent. Clock-face phrasing ("11 o'clock, 2 metres") is used only when ToF is valid, following the study finding that precise phrasing was preferred [2][25]. DESIGN

## 8.3 Haptic mapping

| Event | Left motor | Right motor |
|---|---|---|
| Obstacle left / right | Pulse at band rate / off | Off / pulse at band rate |
| Obstacle centre | Both pulse together at band rate | |

| Event | Left motor | Right motor |
|---|---|---|
| NEAR (any side) | Strong 150 ms buzz on the hazard side(s), then band pulses | |
| System unavailable | Long-short-short on both sides, plus phone vibration | |

Haptics are the *primary* directional channel in single-sided-deafness mode, and a confirming channel otherwise. The BuzzClip result warns that weak, frequent vibration can be confused with cane-tip feedback [2], so intensity and cadence are user-tunable and validated in Gate G9. DESIGN

## 8.4 Cue bank as data

```
cue_bank.json  (shipped to glasses flash; phone references by cue_id)
{ "obstacle_far":  {"wav":"ob2.wav", "period_ms":1000, "haptic":"pulse40"},
  "obstacle_mid":  {"wav":"ob2.wav", "period_ms":400,  "haptic":"pulse60"},
  "obstacle_near": {"wav":"ob3u.wav","period_ms":200,  "haptic":"buzz150","level_step_db":6},
  "person_*":      {"wav":"pe2.wav", ...},
  "system_off":    {"wav":"sys3d.wav","period_ms":5000,"haptic":"lss","both":true} }
```

CHAPTER 9

# Calibration for asymmetric hearing

**Goal.** Both sides of the cue should be *equally perceptible* to this wearer, with this frame position and this output device, without ever exceeding a hard ceiling. This is perceptual matching. It is not a hearing test or a diagnosis, and it does not restore hearing. DESIGN




Figure 9.1. Calibration flow, roughly 2–3 minutes and fully operable with TalkBack and screen-reader gestures.

1. **Fit and device.** The app asks the wearer to seat the frame. It records which output device is in use (glasses transducers or a headset) and the wearer's self-reported weaker side. It warns about ambient noise if the phone microphone reads high.
2. **Per-side detection threshold.** An adaptive 2-down-1-up staircase, which converges near 70.7% correct [57], starts from a quiet level. The wearer answers "heard / not heard" by tapping. It runs in three bands that match the cue timbres (for example 500 Hz, 1 kHz and 2 kHz) and stops after 6–8 reversals. PROPOSED
3. **Presentation level.** Per band and side, the level is threshold plus a sensation level chosen by the wearer between 10 and 20 dB [24], clamped to the ceiling.
4. **Loudness balance.** Alternate the real "obstacle" cue left and right. The wearer nudges a balance control until both sides feel equally loud, which corrects for cue spectrum and for the bone crossing the skull. Repeat once to check consistency.
5. **Verify and choose a mode.** Play five random-side cues and have the wearer name the side.
   - **Stereo mode:** at least 4 of 5 correct and no side at the ceiling.
   - **SSD mode:** one side cannot be detected at the ceiling, or side naming is at chance. Audio becomes identical on both sides (mono). Direction moves to haptics plus spearcons.
   - **Haptic-only mode:** for wearers who prefer it or cannot use the audio.

## 9.1 Profile and safety rules

```
HearingProfile {
  device_id, created_at, fit_note,
  mode: STEREO | SSD | HAPTIC_ONLY,
  ceiling_db,                      # from device characterisation; glasses enforce independently
  bands: [{hz, thr_db_L, thr_db_R, sl_db, offset_db_L, offset_db_R}],
  haptic: {gain_L, gain_R, pattern_set},
  verified: {trials:5, correct:n}
}
```

- **Ceiling first.** No calibration step, slider or urgency step can exceed `ceiling_db`. If the weak side would need more, it is marked unusable. The app does not push more energy into it. DESIGN
- **Re-run every session**, because how the frame couples to the head changes each time it is worn. A 20-second quick check suffices if the previous profile is under 7 days old. PROPOSED

- **Dose.** Track cumulative cue-on time. Cues are short and sparse by design. Until the transducers are measured on an artificial mastoid, Determine makes no claim of equivalence to the WHO and ITU air-conduction limits [4] [5].
- **Professional care.** The app recommends an audiologist for anyone with known hearing loss and never displays results as audiometry.
- **OS features stay reachable.** Android mono audio and balance settings remain available. Determine's profile works alongside them, not as a replacement.

CHAPTER 10

# Safety, privacy and regulatory boundary

## 10.1 Threat and hazard ledger

| Hazard | Mitigation | Residual risk |
|---|---|---|
| Missed obstacle (glass, wire, out of FOV) | "Cannot see" list. Secondary-aid positioning. Cane mode | Inherent. Disclosed |
| False reassurance | No "clear" cue exists. UNAVAILABLE state. TTL on cues | Users may over-trust silence. Address it in training |
| Over-alerting and fatigue | One alert at a time, rate limit, persistence and hysteresis | Tune against measured false-alert rate |
| Masking environmental sound | Short sparse cues, open ears, cue-on duty tracked [52] | Measure in Gate G10 |
| Excess vibration or sound level | Ceiling enforced on both phone and glasses. Session timer | No bone-conduction dose standard available |
| Electrical and thermal | Pocket USB bank, no head-mounted Li-Po in v0, strain relief, temperature in heartbeat | Hackathon-grade wiring. Supervised use only |
| Bystander privacy | Visible recording indicator, no retention, demo signage and consent | Camera on face may still worry bystanders |
| Spoofed cue commands | v0 hotspot is WPA2 with a unique password. Roadmap: authenticated session key | LAN attacker during demo. Low impact, but logged |

## 10.2 Regulatory boundary

21 CFR 886.5900 defines an *electronic vision aid* as a device that translates visual images of objects into tactile or auditory signals for a person with impaired vision or blindness. It is Class I and exempt from 510(k) premarket notification [31]. FACT Marketing Determine as a mobility aid would likely place it in this category. Being exempt from 510(k) does not mean being exempt from every FDA obligation, so get legal review of the remaining duties before any sale. The hackathon prototype is a research demonstration. INFERENCE

## 10.3 Human-subjects boundary

Testing with blind, low-vision or hearing-impaired participants beyond informal feedback SHOULD go through the university's human-subjects process, include an O&M specialist, and have informed consent and a spotter. Every HackTX demo uses a staged indoor course with a sighted spotter. DESIGN

## 10.4 Meta platform rules (if the DAT adapter is used)

Meta's acceptable-use policy prohibits tampering with the capture LED. It allows face or voice identification only for accessibility apps with Meta's prior approval [8]. Determine does no face identification. FACT

CHAPTER 11

# HackTX 26 build plan

## 11.1 Event facts (official site)

- Hacking runs from **8 am Saturday 24 Oct to 5 pm Sunday 25 Oct 2026** on the UT Austin campus, fully in person. Teams have at most 4 people. The event is free and does not reimburse travel [32].
- Rule: **"All work must be done at the event. You can't demo something you didn't build."** Attendees follow the MLH Code of Conduct [32].
- Prizes exist for challenges, with details "revealed at the opening ceremony". No judging rubric, tracks or prize values are published [32].
- The listed sponsors include Meta (highlighted tile), Visa, Annapurna (linked to Annapurna Labs), NMC2, TFS, BHW, D. E. Shaw and Capital One [32].
- Conflict on the site: the application banner says "Apply by Sept 15" while the FAQ says Sept 11. Walk-in admission is first come, first served, is not guaranteed, and the site discourages non-Austin attendees from travelling for it [32]. **Confirm acceptance in the portal before travelling.**

> **Implication:** all work, including this document, is done at the event: buy parts, read documentation, plan, write firmware, write the app and assemble the frame there. Email `hello@freetailhackers.com` to confirm whether cue sound files and phone compatibility checks are allowed. Get the answer in writing.

## 11.2 Team roles (4 people)

| Role | Owns |
|---|---|
| Firmware | ESP32-CAM JPEG stream; XIAO ESP32-C3 ToF, UDP protocol, I²S cue player, haptic driver, watchdog, heartbeat |
| Android | FrameSource and RangeSource ingest, MediaPipe detector, fusion, HazardPolicy, latency logging |
| Audio / UX | Cue bank, calibration flow, TalkBack-accessible UI, spearcons, judge diagnostics overlay |
| Hardware / demo | Frame mounting, wiring, power, the staged course, test protocol, pitch and Devpost |

## 11.3 Timeline with abort branches

| Window | Milestone | Abort or branch |
|---|---|---|
| Sat 8–12 | Two-board bring-up: ESP32-CAM JPEG reaches the phone over the hotspot; C3 ToF grid streams; two-board power distribution checked; pin maps fixed; cue player plays a stored WAV on both sides | Camera module lacks PSRAM or has an unsupported sensor → drop resolution (CIF-or-lower JPEG needs no PSRAM [13]), else Tier 0 (phone camera on headband) while debugging. Transducers missing → phone-to-headset fallback |
| Sat 12–18 | End-to-end loop: ToF corridor → policy → glasses cue. Detector labels added. Timestamps logged | Detector slow → ToF-only hazard with generic "obstacle" label |
| Sat 18–24 | Calibration flow, three modes, haptics, UNAVAILABLE handling | DRV2605L trouble → MOSFET PWM |
| Sun 0–8 | Gates G6–G8 on the staged course. Tune bands and rates. Freeze features | No new features after 8 am |

| Window | Milestone | Abort or branch |
|---|---|---|
| Sun 8–12 | Optional adapters: Meta DAT through Mock Device Kit, Auracast demo broadcast (only on supported phones) | Skip if any core gate is failing |
| Sun 12–17 | Pitch rehearsal, Devpost, demo resets, spare batteries | — |

### 11.4 Three-minute judge demo

1. **Hook (20 s):** the parking-sensor line. Hold up the two-board frame next to the costs of NOA ($4,990) and OrCam ($4,250).
2. **Live course (75 s):** a sighted spotter walks the wearer past a head-height foam bar, a person, and a left-side then right-side obstacle. The judge diagnostics screen shows frame age, ToF grid, chosen alert and measured latency.
3. **Hearing (40 s):** switch the profile to SSD mode live. The urgent cue stays audible and direction moves to haptics plus spearcons. Show the ceiling stopping a forced gain increase.
4. **Failure (25 s):** unplug the camera. The system-off cue plays within the measured bound, and no "clear" is ever heard.
5. **Evidence (20 s):** the logged p50/p95 latency, the miss and false-alert table, and the BOM.

### 11.5 Sponsor alignment, without forcing it

**Meta**

Wearables DAT 1.0 (rollout from 30 Sep 2026) gives mobile apps camera, microphone, audio and IMU access on supported AI glasses. Its Mock Device Kit simulates the glasses camera without hardware [6]. Determine's `FrameSource` seam lets a DAT adapter feed the *same* hazard loop. The story: Determine runs on commodity dumb glasses *and* on Ray-Ban Meta. Claim it only if the adapter was built at the event.

**Annapurna Labs (AWS)**

Annapurna Labs is AWS's Austin-based chip group behind Nitro, Graviton, Inferentia and Trainium [33]. The honest connection is *system-first co-design*: Determine places computation in the frame, the phone or the cloud according to latency, power and failure budgets (Chapter 6), and measures them. The optional Describe loop targets that silicon: a Graviton2 Lambda API holds the prompt, and SmolVLM would run on Inf2 via Optimum Neuron, with Trainium2 the documented alternative (§7.3) [34][35][62]–[65]. They stay out of the hazard loop.

**Winning overall** is an ambition, not something this document can engineer. What it can engineer is the set of things judges reward: a complete working loop, a clear human need backed by evidence, measured numbers, honest limits and a memorable demo.

CHAPTER 12

# Sourcing in Houston and bill of materials

## 12.1 Verified Houston-area stores

| Store | Address | Hours / phone | What to look for |
|---|---|---|---|
| **Micro Center Houston** | 5305 S Rice Ave, Houston, TX 77081 | Mon–Sat 10–9, Sun 11–6 · (713) 940-8500 [38] | Selected C3, ESP32-CAM, amps, haptic drivers, motors, mux, regulator and FTDI programmer; exact SKUs in §12.2. User reports the listings in stock; shelf counts and reservations were not independently observed |
| **Electronic Parts Outlet (EPO)** | 3753-B Fondren Rd, Houston, TX 77063 | Mon–Fri 9–6, Sat 9–5, Sun closed · (713) 784-0140 [39] | Says it stocks Arduino, SparkFun, Raspberry Pi, Pololu and Adafruit lines. Call about specific parts |
| **Ace Electronics** | 3210 Antoine Dr, Houston, TX 77092 | Mon–Fri 8–5 only · 713-688-8114 [40] | Vibration motors, wire, protoboard, heat shrink, MOSFETs |
| Best Buy Galleria | 5133 Richmond Ave, Houston, TX 77056 | (713) 552-1512 [41] | Commercial bone-conduction headset (fallback audio). Check stock online first |
| **Micro Center Austin** | 12707 N Mopac Expy, Austin, TX 78727 | Opens 10 am Saturday [38] | Emergency top-up during HackTX (hacking starts at 8 am) |

No electronics supplier in Galveston was verified. The user reports the Houston-listed parts in stock; this document has no independent reservation or checkout evidence. Micro Center automated pages blocked inventory reads. EPO lists general components, not per-part stock. The VL53L5CX Mini and raw bone transducers have no verified Houston walk-in source: shipped orders are exceptions, and an “in stock” label alone does not prove a delivery date.

## 12.2 Complete purchase list, compact two-board prototype

| Part and role | Buy | Source, status and constraint |
|---|---|---|
| XIAO ESP32-C3 cue board, 3-pack (one used, two spare) | 1 pack | Micro Center Houston SKU 069005; user-reported in stock |
| Inland ESP32-CAM camera modules, pair (one used, one spare) | 1 pair | Micro Center SKU 221739; user-reported in stock. Check the actual sensor, PSRAM and 5 V pinout at G0 [13] |
| VL53L5CX Qwiic Mini 8×8 ToF, front-facing | 1 | SparkFun SEN-19013, $25.95 when checked 18 Oct; shipped exception. 12.7 × 25.4 mm, not three bulky Modulino boards [20] |
| MAX98357A mono I²S amp, one left and one right | 2 | Micro Center SKU 007112; user-reported in stock. Set separate channel-selection voltages [36] |

| Part and role | Buy | Source, status and constraint |
|---|---|---|
| Adafruit 1674 bone-conduction transducer, 8 Ω / 1 W | 2 | DigiKey 1528-1948-ND; shipped exception unless exact local parts are confirmed. Adafruit direct was out of stock; limit amp output and insulate for skin contact [36] |
| DRV2605L haptic motor driver | 2 | Micro Center SKU 008607; user-reported in stock. Inspect connector style before choosing leads |
| ERM vibration disc motor | 2 | Micro Center SKU 850917; user-reported in stock |
| TCA9548A I²C multiplexer for the two haptic drivers | 1 | Micro Center SKU 278499; user-reported in stock |
| LM3671 3.3 V / 600 mA buck regulator for peripheral rail | 1 | Micro Center SKU 009191; user-reported in stock. Verify measured current under 600 mA |
| FTDI USB-to-UART adapter for ESP32-CAM programming | 1 | Micro Center SKU 489252; user-reported in stock. Use 3.3 V UART logic; obtain its matching computer-side USB cable |
| Qwiic-to-bare-wire 4-pin cable for the Mini ToF | 1 | SparkFun PRT-17912; ship with the Mini, or obtain a compatible local harness |
| 5 V USB power bank with at least two simultaneous outputs | 1 | Local general electronics retailer; verify measured combined draw and both ports in use. Battery stays in pocket |
| USB-C data/power cable for C3; computer-side FTDI USB cable; insulated USB-to-5 V/GND breakout for camera, amps and regulator | 1 each | Buy locally, matching the actual FTDI connector and power-bank output; keep two separately powered branches and common signal ground |
| Jumper wires, 0.1-inch headers, hookup wire, perfboard, solder, heat-shrink and strain relief | 1 small set each | Micro Center or Electronic Parts Outlet; no item-level stock proof. Include the ESP32-CAM flash jumper |
| Resistor assortment and recording indicator LED | 1 set + 1 LED | Local electronics store. Amp channel selection and LED current limiting require resistors; confirm a usable indicator GPIO at G0 |
| Headband or glasses frame, camera/ToF mounts, transducer/motor mounts, Velcro or tape, skin-safe insulation | 1 set | Local hardware or eyewear store. Only camera and Mini ToF face forward; insulate skin-facing transducer joints |
| Android phone for hotspot, offline perception and app | 1 | User-owned if compatible. Soldering iron and multimeter: one each if not already owned |

**No grand total:** prices, local pickup receipts, overnight checkout dates, shipping, tax, power draw and mount fit are not all confirmed. An online “in stock” listing is not a delivery guarantee. The two essential shipped exceptions are the Mini ToF and raw bone transducers; no verified in-person Houston equivalents preserve their functions.

## 12.3 Tiers

| Tier | Contents | Cost status | Use |
|---|---|---|---|
| Tier 0: phone-only | Android phone on a headband using its own camera, plus available headphones and phone vibration. Same app, using the PhoneCamera FrameSource | Unquoted | Software proof; not the glasses |
| Tier 1: minimum frame | C3 cue board + ESP32-CAM + ToF + 2 motors (MOSFET drive) + supplies + frame + power. Audio from phone to existing headphones | Unquoted | Camera and haptic demo |

| Tier | Contents | Cost status | Use |
|---|---|---|---|
| **Tier 2: recommended** | Complete table above: two boards, one compact 8×8 Mini ToF, stereo bone cues played on the cue board, haptics and pocket power | Unquoted | Full staged demo; two shipped exceptions |
| Fallback audio | Replace DIY transducers and amps with a bone-conduction headset | Price at purchase | Haptics carry urgency |

Other listed products (24 Oct 2026): Ray-Ban Meta Gen 3 from $449 [42], Envision Ally Solos $699 [43], OrCam MyEye 3 Pro $4,250 [44], NOA $4,990 [46], .lumen €9,999 [47]. They do different jobs. Determine's two-board total is unquoted, so no like-for-like cost-of-entry comparison is claimed.

## 12.4 Houston-first purchase and size gates

1. **Collect locally:** reserve the user-reported in-stock Micro Center Houston SKUs in §12.2, then check the ESP32-CAM PSRAM/sensor, driver connector revisions and the actual FTDI port. Buy generic wiring and mounts from Micro Center or EPO. The user report is not a receipt or a measured shelf count.
2. **Two shipped exceptions:** order one SparkFun VL53L5CX Mini plus its Qwiic harness, and two Adafruit 1674 transducers from DigiKey unless the exact items are found locally. Confirm delivery dates at checkout; choose expedited service if required.
3. **Before wiring:** mock up a single forward camera and ToF on a headband, with control boards on the rear strap and battery in a pocket. If balance, fit or simultaneous power fails Gate G0, do not claim a wearable build. A single locally sold Modulino Distance board would be a larger, 1.2 m single-zone near-field demo only, not an equivalent 8×8 fallback.

CHAPTER 13

# Verification gates and failure catalogue

| Gate | Test | Pass criterion (PROPOSED unless stated) |
|---|---|---|
| G0 | Bring-up on two boards: ESP32-CAM JPEG frames and cue-board ToF at 15 Hz, with both pin maps and the two-board power distribution fixed. Record the camera module's actual PSRAM and sensor | 10 minutes, no resets, no corrupt JPEGs, no brown-outs when both radios transmit |
| G1 | Both Wi-Fi producers stream to the team's phone over one hotspot simultaneously | ≥ 10 fps QVGA/VGA JPEG sustained 10 min with the ToF stream active. Per-board frame and range loss plus staleness logged |
| G2 | USB UVC on the team's phone (optional path) | Enumerates via Camera2 or a user-space library. Otherwise mark it unsupported |
| G3 | Cue path: phone command → glasses earcon, both sides | True L/R separation confirmed. p95 command-to-sound measured |
| G4 | Gain ceiling enforced on the glasses | A forged CUE above the ceiling is clamped |
| G5 | Calibration repeatability | Two runs within ±3 dB per side, same fit |
| G6 | End-to-end latency on the course | p50/p95/max over ≥ 100 events. Target p95 ≤ 250 ms (report the actual value either way) |
| G7 | Fault injection: unplug or power down either board, kill app, Wi-Fi off, low battery | Unavailable cue ≤ 1 s. No stale cue after the fault |
| G8 | Detection on the staged course: head-height bar, person, left and right obstacles, 10 repeats each | Miss and false-alert table published. No "works" claim without it |
| G9 | Cue identification after 5 minutes of training | About 90% correct family and side, in ambient noise (the number is arbitrary and needs a pilot) |
| G10 | Masking check: localise real sounds with cues on versus off | Report the difference. Reduce cue duty if it is large [52] |
| G11 | Thermal and battery over 30 minutes | Frame touch temperature comfortable. Runtime measured |

## 13.1 Failure catalogue

| Failure | Detection | Response |
|---|---|---|
| Phone lacks external-camera HAL | G2 | Wi-Fi path (default anyway) |
| ESP32-CAM module has no PSRAM or an unsupported sensor | G0 bring-up; module markings vs the esp32-camera lists [13] | The driver requires PSRAM above CIF-or-lower JPEG [13]: drop resolution, else Tier 0 phone camera |
| PSRAM contention corrupts frames under Wi-Fi [13] | JPEG decode errors | Lower resolution or frame rate. Use JPEG only |
| Hotspot drops or the phone roams to another network | Heartbeat timeout on both boards | UNAVAILABLE, then auto-rejoin. Pin the SSID |
| One board drops off the hotspot while the other stays up | That board's BEAT times out | Fail loud: UNAVAILABLE, never a half-working "all clear". Auto-rejoin |

| Failure | Detection | Response |
|---|---|---|
| Clock-offset estimate drifts; a stale frame or grid arrives | Freshness gate (offset-corrected age + receive age) | Discard the input, re-estimate from PING/PONG. Persistent staleness → UNAVAILABLE |
| ToF null zones in sunlight or on glass | Status flags | DEGRADED announcement. Camera-only labels |
| Detector misses an unusual obstacle | G8 log | The ToF corridor still triggers "obstacle" |
| A2DP latency in fallback audio [16] | G3 measurement | Haptics carry urgency. Audio carries labels |
| Headset multipoint switches to another device | Audio route change callback | Alert on phone. Disable multipoint for the demo |
| Wearer cannot lateralise bone cues | Calibration verify step | SSD mode: mono audio, haptic and spoken direction |
| Brown-out when both boards transmit at once | Reset reason in heartbeat | Bulk capacitance, short separate power runs, power bank with adequate output |

CHAPTER 14

# Roadmap after HackTX

| Track | Next step | Gate before claiming |
|---|---|---|
| User research | Co-design sessions with blind and low-vision travellers, O&M specialists, and people with unilateral hearing loss | Ethics approval. No blindfolded substitutes [2] |
| Hearing-aid output | Send cues directly to LE Audio hearing aids and earbuds (unicast, LC3), so wearers who already use hearing aids need no extra transducer [16][19] | Device support plus measured latency |
| Auracast | Encrypted broadcast so an O&M instructor or companion can monitor cues in training [17][18] | Supported phones only. Never the safety path |
| Meta DAT | Production DAT adapter on supported Ray-Ban and Oakley Meta models, with bone and haptic output kept on a separate accessory [6][7] | Measured stream latency (undocumented by Meta) |
| Sensing | Synchronised stereo or ESP32-P4 with high-speed USB. A downward-pitched sensor for drop-off *research* | Per-class recall data with O&M review |
| Hardware | Custom flex PCB in the temple arms, integrated battery with protection, IP rating | Electrical, battery and RF compliance testing |
| Single-board wearable | Re-combine camera and cues on one SoC when a camera-capable board (e.g. XIAO ESP32-S3 Sense [14]) is reliably obtainable; removes the second radio and the per-board clock alignment | Availability plus a G0-equivalent two-function bring-up |
| iOS | Wi-Fi FrameSource. Apple documents UVC external cameras for iPad [56], so treat iPhone USB cameras as unsupported until tested | Parity of gates G1–G7 |
| Regulatory | Classification review under 21 CFR 886.5900 [31] | Legal counsel |

CHAPTER 15

# Architecture decision records

| ADR | Decision | Alternatives rejected | Revisit when |
|---|---|---|---|
| 001 | Phone is the only decision-maker | On-glasses inference (cost, heat, battery). Cloud inference (latency, offline failure) | A frame SoC can run the policy at similar cost and power |
| 002 | Mono RGB + 8×8 ToF for v0 | Dual RGB stereo (sync and calibration). Ultrasonic (wide beam, weak bearing) | Synchronised stereo is affordable and measured outdoors |
| 003 | Wi-Fi hotspot transport for frames | BLE video (bandwidth). USB UVC by default (support varies by phone maker) | Gate G2 passes on the target phone fleet |
| 004 | Urgent cues played on the glasses from stored earcons | A2DP streaming (100–200 ms). Phone speaker (no direction, public) | LE Audio unicast is widely available |
| 005 | Direction is always redundant; no panning-only cues | HRTF spatial audio alone (4 of 13 struggled [2]; SSD cannot use it) | Never for SSD mode |
| 006 | Calibration matches perceived level per session, with a hard ceiling enforced twice | Fixed balance slider. Unlimited boost of the weak side | A transducer dose standard exists |
| 007 | No "clear" cue; silence means no report from a healthy system | Periodic "all clear" reassurance | Never |
| 008 | Cloud and generative description out of the hazard loop | LLM-driven navigation (22.2% wrong answers [30]) | Never for hazard cues |
| 009 | Meta DAT is a FrameSource adapter, not a dependency | DAT-only build (cost, no haptics or bone output) | The team owns supported glasses |
| 010 | Android first | iOS first (no UVC, Classic audio only) | After gates G1–G7 pass on Android |
| 011 | Two Wi-Fi producers: XIAO ESP32-C3 cue board + separate ESP32-CAM camera board, no board-to-board bus (24 Oct 2026) | Single XIAO ESP32-S3 Sense (not obtainable for this build). Hardwired board-to-board bus (unverified wiring, no spare bus on the C3, new failure modes) | A camera-capable single board is reliably purchasable, or two-station hotspot throughput fails Gate G1 |

CHAPTER 16

# Source register

Retrieved 24 Oct 2026, except [13], [20], [36], [59] and the AWS entries re-read for the 24 Oct two-board revision. Only the SparkFun ToF and Adafruit bone transducer price/stock were rechecked among the older-priced parts. Vendor pages state vendor claims. Product prices drift.


1. Lighthouse Guild, "The White Cane and Beyond" (2024). https://lighthouseguild.org/news/the-white-cane-and-beyond/
2. Pittet et al., secondary ETA evaluation with BVI participants, *Scientific Reports* (2026). https://pmc.ncbi.nlm.nih.gov/articles/PMC12909938/
3. Manduchi & Kurniawan, mobility-related accidents of blind travellers (2011). https://users.soe.ucsc.edu/~manduchi/papers/MobilityAccidents.pdf
4. WHO, "Deafness and hearing loss: Safe listening" (6 Mar 2026). https://www.who.int/news-room/questions-and-answers/item/deafness-and-hearing-loss-safe-listening
5. WHO-ITU, Global standard for safe listening devices and systems. https://www.who.int/publications/i/item/9789241515276
6. Meta, Wearables developer FAQ (DAT 1.0, supported devices, Mock Device Kit). https://developers.meta.com/wearables/faq/
7. Meta, DAT microphones and speakers. https://wearables.developer.meta.com/docs/develop/dat/microphones-and-speakers/
8. Meta, Wearables acceptable-use policy. https://wearables.developer.meta.com/acceptable-use-policy
9. AFB AccessWorld, Meta glasses review (Fall 2025). https://afb.org/aw/fall2025/meta-glasses-review
10. AOSP, External USB cameras. https://source.android.com/docs/core/camera/external-usb-cameras
11. Android Developers, USB host. https://developer.android.com/develop/connectivity/usb/host
12. Espressif, ESP32-S3 datasheet. https://documentation.espressif.com/esp32-s3_datasheet_en.pdf
13. Espressif, esp32-camera. https://github.com/espressif/esp32-camera
14. Seeed Studio, XIAO ESP32S3 Sense. https://www.seeedstudio.com/XIAO-ESP32S3-Sense-p-5639.html
15. Espressif FAQ, BLE throughput. https://docs.espressif.com/projects/esp-faq/en/latest/software-framework/bt/ble.html
16. Bluetooth SIG, *Introducing Bluetooth LE Audio*. https://www.bluetooth.com/wp-content/uploads/2022/01/Introducing-Bluetooth-LE-Audio-book.pdf
17. Bluetooth SIG, How to build an Auracast transmitter (rev. 16 Sep 2026). https://www.bluetooth.com/wp-content/uploads/2024/05/2403_How_To_Auracast_Transmitter.pdf
18. Google, LE Audio and Auracast support on Android (2025). https://blog.google/products-and-platforms/platforms/android/le-audio-auracast-support/
19. Android Developers, BLE Audio overview. https://developer.android.com/develop/connectivity/bluetooth/ble-audio/overview
20. SparkFun, Qwiic Mini ToF Imager VL53L5CX. https://www.sparkfun.com/sparkfun-qwiic-mini-tof-imager-vl53l5cx.html
21. Google AI Edge, MediaPipe Object Detector for Android. https://ai.google.dev/edge/mediapipe/solutions/vision/object_detector/android
22. Google AI Edge, LiteRT for Android. https://ai.google.dev/edge/litert/android
23. Depth Anything V2 (licences per model size). https://github.com/DepthAnything/Depth-Anything-V2
24. Brewster, Wright & Edwards, earcon design guidelines (1995). https://www.dcs.gla.ac.uk/~stephen/papers/HCI95.pdf
25. Walker et al., spearcons (2013). https://pubmed.ncbi.nlm.nih.gov/23516800/
26. Stenfelt, transcranial attenuation in unilateral deafness (2012). https://pubmed.ncbi.nlm.nih.gov/22193619/
27. Stanley & Walker, bone-conduction lateralisation (2006). https://sonify.psych.gatech.edu/publications/pdfs/2006HFES-StanleyWalker.pdf
28. Agterberg et al., CROS bone-conduction devices and localisation (2019). https://pubmed.ncbi.nlm.nih.gov/29703651/
29. Van Wanrooij & Van Opstal, monaural localisation (2004). https://pubmed.ncbi.nlm.nih.gov/15115811/
30. Gonzalez et al., multimodal LLM diary study (2026). https://arxiv.org/abs/2602.13469
31. 21 CFR 886.5900, Electronic vision aid. https://www.law.cornell.edu/cfr/text/21/886.5900
32. Freetail Hackers, HackTX 26 official site and its published bundle (FAQ, sponsors). https://hacktx.com/
33. About Amazon, inside Annapurna Labs. https://www.aboutamazon.com/news/aws/take-a-look-inside-the-lab-where-aws-makes-custom-chips
34. AWS, Inferentia. https://aws.amazon.com/ai/machine-learning/inferentia/
35. AWS, Neuron SDK. https://aws.amazon.com/ai/machine-learning/neuron/
36. Adafruit products: DRV2605L https://www.adafruit.com/product/2305 · motor disc https://www.adafruit.com/product/1201 · MAX98357A https://www.adafruit.com/product/3006 · bone transducer https://www.adafruit.com/product/1674
37. Shokz, OpenMove https://shokz.com/products/openmove · OpenRun https://shokz.com/products/openrun
38. Micro Center, store locations and hours. https://microcenter.zendesk.com/hc/en-us/articles/24790484266007-Micro-Center-Store-Locations-and-Addresses
39. Electronic Parts Outlet. https://epohouston.com/

40. Ace Electronics. https://ace4parts.com/
41. Best Buy Galleria store page. https://stores.bestbuy.com/tx/houston/5133-richmond-ave-291.html
42. Meta store, Ray-Ban Meta. https://www.meta.com/ai-glasses/ray-ban-meta/
43. Envision, Ally Solos shop. https://shop.letsenvision.com/products/ally-solos-glasses
44. OrCam MyEye 3 Pro (reseller price). https://www.getorcam.com/product-page/myeye-3-pro
45. WeWALK product. https://wewalk.io/en/product
46. NOA (reseller). https://floridareading.com/products/noa
47. .lumen glasses. https://www.dotlumen.com/glasses
48. NaviLens. https://www.navilens.com/
49. Microsoft Seeing AI. https://www.microsoft.com/en-us/garage/wall-of-fame/seeing-ai/
50. Google Lookout help. https://support.google.com/accessibility/android/answer/9031274
51. Microsoft Soundscape. https://www.microsoft.com/en-us/research/product/soundscape/
52. May & Walker, bone-conduction audio and environmental localisation (2017). https://pubmed.ncbi.nlm.nih.gov/28237013/
53. Raspberry Pi Zero 2 W product brief. https://pip-assets.raspberrypi.com/categories/584-raspberry-pi-zero-2-w/documents/RP-008359-DS-1-raspberry-pi-zero-2-w-product-brief.pdf
54. Raspberry Pi Pico 2 W datasheet. https://pip-assets.raspberrypi.com/categories/1088-raspberry-pi-pico-2-w/documents/RP-008304-DS-3-pico-2-w-datasheet.pdf
55. Espressif, ESP32-P4. https://www.espressif.com/en/products/socs/esp32-p4
56. Apple, AVCaptureDevice external device type. https://developer.apple.com/documentation/avfoundation/avcapturedevice/devicetype-swift.struct/external
57. Levitt, transformed up-down methods in psychoacoustics (1971). https://doi.org/10.1121/1.1912375
58. Meta, DAT StreamConfiguration reference (v0.9). https://wearables.developer.meta.com/docs/reference/android/dat/0.9/com_meta_wearable_dat_camera_types_streamconfiguration
59. Seeed Studio, XIAO ESP32-C3 Getting Started (400 KB SRAM, 4 MB flash, 1×I2C, 1×SPI, 11 GPIO, no integrated camera, typical 75 mA Wi-Fi-active). https://wiki.seeedstudio.com/XIAO_ESP32C3_Getting_Started/
60. AWS Neuron, NxD Inference release notes (Trn2 and later). https://awsdocs-neuron.readthedocs-hosted.com/en/latest/release-notes/components/nxd-inference.html
61. AWS Neuron, Qwen2-VL on Trainium2 tutorial. https://awsdocs-neuron.readthedocs-hosted.com/en/latest/libraries/nxd-inference/tutorials/qwen2-vl-tutorial.html
62. Hugging Face, Optimum Neuron supported architectures. https://huggingface.co/docs/optimum-neuron/en/supported_architectures
63. AWS, EC2 Inf2 instances. https://aws.amazon.com/ec2/instance-types/inf2/
64. AWS, EC2 Trn2 instances. https://aws.amazon.com/ec2/instance-types/trn2/
65. AWS, Lambda arm64 / Graviton2 foundations. https://docs.aws.amazon.com/lambda/latest/dg/foundation-arch.html
66. AWS, Bedrock Nova Lite model card. https://docs.aws.amazon.com/bedrock/latest/userguide/model-card-amazon-nova-lite.html
67. AWS, EC2 accelerator instance quotas. https://docs.aws.amazon.com/ec2/latest/instancetypes/ec2-instance-quotas.html